

RMC-core — Timing-Constraint Reference

Every JEDEC delay, its arb-level scope, the command it gates, and the datapath-busy budget.

Speed grade **DDR5-4800B**, $t_{CK} = 0.4167$ ns, BL16 (one 64 B packet = one CAS). All values in **CK** unless noted; ns = CK $\times$ 0.4167.

Source: JEDEC JESD79-5 AC timing (Table 252 for turnaround) and the DDR5-4800B bin. Grade **4800B** = **40-40-40**:

$t_{RCD} = t_{RP} = 40$, $t_{RC} = 117$, $t_{RAS} = 77$, CWL = 38. Command-pair structure from DDR4_Command_Timing.csv, values substituted with DDR5-4800B. (An earlier draft used 34/34/111 from the Xilinx DDR5_timing_params.csv row — that is the **4800A** grade / absolute JEDEC min, not 4800B; reverted to the 4800B bin here.)

1 Gate model (recap)

Every command a bank can issue is legal only when its **4 folded 1-bit gates** are all high. Each timing parameter is owned by exactly one *scope level* and drives one gate at that level; the resident gate the eligibility logic reads is the AND across levels:

$$\text{can_x} = \text{can_x_L0} \& \text{can_x_L1} \& \text{can_x_L2} \& \text{can_x_L3}, \quad x \in \{\text{act, pre, rcas, wcas}\}.$$

Each level bit is a comparator $\text{can_x_Ln} = (\text{GC} \geq \text{next_x_Ln})$, where $\text{next_x} = \text{GC}_{\text{issue}} + t$. Scopes:

Level	scope	instances	tracks
L0	same bank	$N_{\text{rank}} \times 32 = 64$	this bank's own last op
L1	same bank-group	$N_{\text{rank}} \times 8 = 16$	any sibling bank in the BG
L2	diff-BG, same rank	$N_{\text{rank}} = 2$	rank-wide last op + tFAW window
L3	diff rank	1 (global)	DQ-bus rank handoff

2 Master parameter table

Value @ DDR5-4800B. **Level** = where the constraint is enforced. **Gates** = which **can_*** it feeds.

Param	CK	ns	Level	Gates	Meaning (from → to)
tRCD	40	16.7	L0	rcas, wcas	ACT → CAS: row-to-column, sense done
tRP	40	16.7	L0	act	PRE → ACT: row precharged, bank re-open-able
tRAS	77	32.1	L0	pre	ACT → PRE: min row-open time
tRC	117	48.8	L0	act	ACT → ACT same bank (=tRAS+tRP)
tRTP	18	7.5	L0	pre	RD → PRE: read-to-precharge
tWR	72	30.0	L0	pre	WR(last beat) → PRE: write-recovery, cell restore
tRRD_L	12	5.0	L1	act	ACT → ACT same BG
tCCD_L	12	5.0	L1	rcas	RD → RD same BG
tCCD_L_WR	48	20.0	L1	wcas	WR → WR same BG (the write penalty)
tWTR_L	24	10.0	L1	rcas	WR → RD same BG turnaround
tRRD_S	8	3.3	L2	act	ACT → ACT diff BG
tCCD_S	8	3.3	L2	rcas, wcas	CAS → CAS diff BG (= DQ heartbeat)
tWTR_S	6	2.5	L2	rcas	WR → RD diff BG turnaround
tRTW	12	5.0	L2	wcas	RD → WR turnaround (rank-wide, no S/L split)
tFAW	32	13.3	L2	act	rolling 4-ACT window per rank
tPPD	2	0.8	L2	pre	PRE → PRE rank power budget
tRTRS	2	0.8	L3	rcas, wcas	CAS → CAS diff rank: DQ-bus handoff
<i>Data-phase latencies (deterministic shift lines, not gates):</i>					
CL	40	16.7	—	—	RD CAS → first read beat on DQ (read latency)
CWL	38	15.8	—	—	WR CAS → first write beat on DQ (write latency)
BL16 burst	8	3.3	—	—	8 CK on DQ per CAS (=BL/2), the busy window
<i>Maintenance (maintenance-engine owned, not in the arb gates):</i>					
tRFC1	708	295	ME	gate_rfc	REFab recovery (all-bank refresh)
tRFCsb	312	130	ME	gate_rfc	REFsb recovery (same-bank refresh)

Param	CK	ns	Level	Gates	Meaning (from → to)
tREFI	9360	3900	ME	—	average refresh interval (debt counter)
tMOD	36	15.0	ME	—	mode-register update settle
tMRD	16	6.7	ME	—	MRW-to-MRW

3 Per-level gate equations

3.1 L0 — same bank (per cell ×64)

$$\begin{aligned}
\text{can_act_L0} &= (\text{GC} \geq \text{lastPRE} + t_{RP}) \& (\text{GC} \geq \text{lastACT} + t_{RC}) \\
\text{can_pre_L0} &= (\text{GC} \geq \text{lastACT} + t_{RAS}) \& (\text{GC} \geq \text{lastRD} + t_{RTP}) \& (\text{GC} \geq \text{lastWR} + t_{WR}) \\
\text{can_rcas_L0} &= (\text{GC} \geq \text{lastACT} + t_{RCD}) \& (\text{GC} \geq \text{lastCAS} + t_{CCD_L}) \& (\text{GC} \geq \text{lastWR} + t_{WTR_L}) \\
\text{can_wcas_L0} &= (\text{GC} \geq \text{lastACT} + t_{RCD}) \& (\text{GC} \geq \text{lastCAS} + t_{CCD_L_WR}) \& (\text{GC} \geq \text{lastRD} + t_{RTW})
\end{aligned}$$

3.2 L1 — same bank-group (per BG ×16, broadcast to its 4 banks)

$$\begin{aligned}
\text{can_act_L1} &= \text{GC} \geq \text{lastACT}_{bg} + t_{RRD_L} \\
\text{can_pre_L1} &= 1 \quad (\text{no same-BG PRE rule}) \\
\text{can_rcas_L1} &= (\text{GC} \geq \text{lastRD}_{bg} + t_{CCD_L}) \& (\text{GC} \geq \text{lastWR}_{bg} + t_{WTR_L}) \\
\text{can_wcas_L1} &= \text{GC} \geq \text{lastWR}_{bg} + t_{CCD_L_WR}
\end{aligned}$$

3.3 L2 — diff-BG, same rank (per rank ×2)

$$\begin{aligned}
\text{can_act_L2} &= (\text{GC} \geq \text{lastACT}_{rk} + t_{RRD_S}) \& (\text{GC} \geq \text{faw_ts}[\text{oldest}] + t_{FAW}) \\
\text{can_pre_L2} &= \text{GC} \geq \text{lastPRE}_{rk} + t_{PPD} \\
\text{can_rcas_L2} &= (\text{GC} \geq \text{lastCAS}_{rk} + t_{CCD_S}) \& (\text{GC} \geq \text{lastWR}_{rk} + t_{WTR_S}) \\
\text{can_wcas_L2} &= (\text{GC} \geq \text{lastCAS}_{rk} + t_{CCD_S}) \& (\text{GC} \geq \text{lastRD}_{rk} + t_{RTW})
\end{aligned}$$

3.4 L3 — diff rank (global ×1)

$$\begin{aligned}
\text{can_act_L3} &= 1 \quad \text{can_pre_L3} = 1 \quad (\text{cross-rank ACT/PRE free}) \\
\text{can_rcas_L3} = \text{can_wcas_L3} &= \underbrace{(\text{GC} \geq \text{dqFree_ts})}_{\text{DQ occupancy}} \& \underbrace{[(\text{rank} = \text{last_cas_rank}) ? 1 : (\text{GC} \geq \text{last_cas_ts} + t_{RTRS})]}_{\text{bus handoff}}
\end{aligned}$$

where $\text{dqFree_ts} = \text{last_cas_ts} + \text{BL}/2$ (8), written by the packer at CAS-commit. **dqFree is the binding cross-rank floor:** same-rank CAS already pay $t_{CCD_S} = 8 \geq \text{burst}$, but a rank hop pays only $t_{RTRS} = 2 < 8$ — tRTRS covers the electrical handoff (ODT/driver), *not* the 8-CK burst occupancy, so without dqFree two ranks would dump overlapping bursts on DQ. A *fresh/idle* CAS has dqFree_ts in the past $\Rightarrow$ gate open, issue now. The RD $\leftrightarrow$ WR *direction-flip* hole is separate (t_{WTR}/t_{RTW} , already carried at L0–L2 rank scope, since there $\text{CL} \neq \text{CWL}$). The CA-bus slot (t_{caFree} , 2-cyc = 2 phases) and the data-engine-free check (WD_SRAM fetch + splitter/accumulator busy = burst = subsumed by dqFree) live at the **packer**, not as selection gates.

4 Inter-command delay matrix

Effective spacing **from** the row command **to** the next, by relationship. This is the fold of the tables above into the numbers a scheduler actually waits (CK).

from	to	same bank	same BG	diff BG	diff rank
ACT	ACT	117 (tRC)	12 (tRRD_L)	8 (tRRD_S)	0 [†]
ACT	CAS	40 (tRCD)	—	—	—
RD	RD	12 (tCCD_L)	12 (tCCD_L)	8 (tCCD_S)	~10 (tCCD_S+tRTRS)
WR	WR	48 (tCCD_L_WR)	48 (tCCD_L_WR)	8 (tCCD_S)	~10
WR	RD	70	70	52	~48 [‡]
RD	WR	12 (tRTW)	12	12	12
RD	PRE	18 (tRTP)	—	—	—
WR	PRE	118 [§]	—	—	—
PRE	ACT	40 (tRP)	—	—	—
PRE	PRE	—	—	2 (tPPD)	0

[†] cross-rank ACT has no tRRD/tFAW — different rank, own budget (the multi-rank parallelism win).

[‡] WR→RD cross-rank *skips* tWTR (write-recovery is internal to the writing rank's array, no hazard for the other rank), but the write burst must still drain off DQ: $\approx \text{CWL} + \text{burst} + \text{tRTRS} = 38 + 8 + 2 = 48$.

[§] WR→PRE = $\text{CWL}(38) + \text{burst}(8) + \text{tWR}(72) = 118$ from the write CAS: the bank cannot close until the array cell finishes restoring.

Derivations: WR→RD same-BG = $\text{CWL} + 8 + \text{tWTR_L} = 38 + 8 + 24 = 70$; diff-BG = $38 + 8 + 6 = 52$.

5 Datapath-busy timing

The DQ bus is the real throughput limiter, not the command bus. One CAS = one BL16 burst = **8 CK on DQ**. Sustained ceiling: $\leq 1 \text{ CAS per } 8 \text{ CK} = t_{\text{CCD_S}}$. Back-to-back diff-BG CAS at tCCD_S = 8 exactly fills the DQ with zero bubble (the DQ heartbeat); same-BG at tCCD_L = 12 leaves a 4-CK gap; same-BG writes at tCCD_L_WR = 48 leave a 40-CK crater.

5.1 Read path (CAS to completion)

CK from RD CAS	event
0	RD CAS issued (col on CA bus)
40	first read beat on DQ (=CL); DQ busy 40..47
~48	last beat captured; RD accumulator has full 64 B
~48	one wide 512-b write into RD_SRAM @ dbuf_addr
~49	rd_done{rob_index, pkt_num} → COMPLETION → resp AFIFO

RL_line is a deterministic CL-deep shift carrying only {rob_index, dbuf_addr}; it self-pops at CL. No CAM. Completion never fires before the final beat is committed to SRAM.

5.2 Write path (CAS to retire) — two separate completion events

CK from WR CAS	event
0	WR CAS issued (col on CA bus)
38	first write beat driven on DQ (=CWL); WD splitter reads 512 b from WD_SRAM
46	last beat driven; DQ free
46	CIF-facing done: wr_done → COMPLETION, wd_slot freed (self-timed, no bus ack)
118	bank retire: 46 + t _{WR} (72) — earliest the bank may PRE

Two distinct events (do not conflate): the CIF completion + slot free at **46** (write is fire-and-forget, controller-generated, no DRAM response), and the array-level retire at **118** (the cell capacitor finished charging). Precharging before 118 disconnects the sense amps mid-restore ⇒ data loss. This is why a same-bank write streak stacks tWR: each different-row write pays the full close cycle.

5.3 Physical write cycle (why same-bank writes cost 198)

WR → PRE = 118, + t_{RP}(40) → ACT @ 158, + t_{RCD}(40) → WR @ 198. So a same-bank *different-row* write cycle = 198 CK; a same-row write is just t_{CCD_L_WR} = 48 (no PRE, tWR paid once at the final close). Read cycle same-bank different-row = t_{RC} = 117.

6 Datapath timing constraints (summary)

- **DQ ceiling:** $\leq 1 \text{ CAS} / 8 \text{ CK}$ sustained. Extra command-bus phases fill with ACT/PRE/REF to *other* banks.
- **Read latency:** data lands CL = 40 CK after RD CAS; RD accumulator does one 512-b SRAM write per packet (no per-beat dribble).
- **Write latency:** data leaves CWL = 38 CK after WR CAS; WR splitter does one 512-b SRAM read per packet.
- **Write retire:** bank locked from PRE until CAS+118; CIF completion decoupled at CAS+46.
- **Turnaround on one DQ bus:** R→W = tRTW = 12; W→R = CWL+8+tWTR (70/52 same/diff-BG), skipped cross-rank (~tRTRS).
- **RL/WL lines carry no timestamp** — fixed shift depth *is* the timing; load at CAS-commit, self-pop at CL/CWL.
- **Gear:** at 1:N, 1 mc_clk = N CK; GC counts CK and advances by N/mc_clk so all deadlines stay CK-granular. Accumulator/splitter are the only gear-aware blocks.

7 Full JEDEC command-to-command matrix (DDR5-4800B)

Every command pair from the JESD79 command matrix (DDR4_Command_Timing.csv structure), evaluated with the DDR5-4800B constants above (CK). **same bank** / **diff bank** / **diff rank** scopes. “—” = illegal in that state; “sys” = system-dependent (max window). Auto-precharge (RDA/WRA), self-refresh (SRE/SRX), power-down (PDE/PDX) and MRS rows are included for completeness — the scheduler uses only the ACT/RD/WR/PRE/REF families; the rest are maintenance-engine / init-path.

Command pair	same bank	diff bank	diff rank
ACT-to-DES	1 tCK	1 tCK	1 tCK
ACT-to-NOP	1 tCK	1 tCK	1 tCK
ACT-BST	NA dep	NA dep	NA dep
ACT-to-RD	40	1 tCK	1 tCK
ACT-to-RDA	40	1 tCK	1 tCK
ACT-to-WR	40	1 tCK	1 tCK
ACT-to-WRA	40	1 tCK	1 tCK
ACT-to-ACT	117	12	1 tCK
ACT-to-PRE	77	1 tCK	1 tCK
ACT-to-PREA	77	77	1 tCK
ACT-to-AREF	117	117	1 tCK
ACT-to-SRE	sys	sys	sys
ACT-to-PDE	sys	sys	sys
ACT-to-MRS	sys	sys	sys
RD-to-DES	1 tCK	1 tCK	1 tCK
RD-to-NOP	1 tCK	1 tCK	1 tCK
RD-to-BST	NA dep	NA dep	NA dep
RD-to-RD	12	8	10
RD-to-RDA	12	8	10
RD-to-WR	12	12	12
RD-to-ACT	58	1 tCK	1 tCK
RD-to-PRE	18	1 tCK	1 tCK
RD-to-PREA	18	18	1 tCK
RD-to-WRA	12	12	12
RD-to-AREF	58	58	1 tCK
RD-to-SRE	sys	sys	sys
RD-to PDE	sys	sys	sys
RD-to-MRS	sys	sys	sys
WR-to-DES	1 tCK	1 tCK	1 tCK
WR-to-NOP	1 tCK	1 tCK	1 tCK
WR-to-BST	NA dep	NA dep	NA dep
WR-to-RD	70	52	48
WR-to-WRA	48	8	10
WR-to-WR	48	8	10
WR-to-ACT	158	1 tCK	1 tCK
WR-to-PRE	118	1 tCK	1 tCK
WR-to-PREA	118	118	1 tCK
WR-to-RDA	100	52	48
WR-to-AREF	158	158	1 tCK
WR-to-SRE	sys	sys	sys
WR-to-PDE	sys	sys	sys
WR-to-MRS	sys	sys	sys
RDA-to-DES	1 tCK	1 tCK	1 tCK
RDA-to-NOP	1 tCK	1 tCK	1 tCK
RDA-to-BST	NA dep	NA dep	NA dep
RDA-to-RD	12	8	10
RDA-to-WR	12	12	12
RDA-to-ACT	58	1 tCK	1 tCK
RDA-to-PRE	NA	1 tCK	1 tCK
RDA-to-PREA	NA	18	1 tCK
RDA-to-RDA	12	8	10
RDA-to-WRA	12	12	12
RDA-to-AREF	58	58	1 tCK
RDA-to-SRE	sys	sys	sys
RDA-to PDE	sys	sys	sys
RDA-to-MRS	sys	sys	sys
WRA-to-DES	1 tCK	1 tCK	1 tCK
WRA-to-NOP	1 tCK	1 tCK	1 tCK
WRA-to-BST	NA dep	NA dep	NA dep
WRA-to-RD	100	52	48
WRA-to-WR	48	8	10
WRA-to-WRA	48	8	10
WRA-to-ACT	158	1 tCK	1 tCK
WRA-to-PRE	NA	1 tCK	1 tCK
WRA-to-RDA	100	52	48
WRA-to-PREA	NA	118	1 tCK
WRA-to-AREF	158	158	1 tCK
WRA-to-SRE	sys	sys	sys

Command pair	same bank	diff bank	diff rank
WRA-to-PDE	sys	sys	sys
WRA-to-MRS	sys	sys	sys
BST-to-DES	NA dep	NA dep	NA dep
BST-to-NOP	NA dep	NA dep	NA dep
BST-to-BST	NA dep	NA dep	NA dep
BST-to-RD	NA dep	NA dep	NA dep
BST-to-WR	NA dep	NA dep	NA dep
BST-to-WRA	NA dep	NA dep	NA dep
BST-to-ACT	NA dep	NA dep	NA dep
BST-to-PRE	NA dep	NA dep	NA dep
BST-to-PREA	NA dep	NA dep	NA dep
BST-to-RDA	NA dep	NA dep	NA dep
BST-to-AREF	NA dep	NA dep	NA dep
BST-to-SRE	NA dep	NA dep	NA dep
BST-to-PDE	NA dep	NA dep	NA dep
BST-to-MRS	NA dep	NA dep	NA dep
PRE-to-DES	1 tCK	1 tCK	1 tCK
PRE-to-NOP	1 tCK	1 tCK	1 tCK
PRE-to-BST	NA dep	NA dep	NA dep
PRE-to-RD	NA	1 tCK	1 tCK
PRE-to-WR	NA	1 tCK	1 tCK
PRE-to-WRA	NA	1 tCK	1 tCK
PRE-to-ACT	40	1 tCK	1 tCK
PRE-to-PRE	1 tCK	2	1 tCK
PRE-to-PREA	2	2	1 tCK
PRE-to-RDA	NA	1 tCK	1 tCK
PRE-to-AREF	40	40	1 tCK
PRE-to-SRE	sys	sys	sys
PRE-to-PDE	sys	sys	sys
PRE-to-MRS	sys	sys	sys
PREA-to-DES	1 tCK	1 tCK	1 tCK
PREA-to-NOP	1 tCK	1 tCK	1 tCK
PREA-to-BST	NA dep	NA dep	NA dep
PREA-to-RD	NA	NA	1 tCK
PREA-to-RDA	NA	NA	1 tCK
PREA-to-WR	NA	NA	1 tCK
PREA-to-WRA	NA	NA	1 tCK
PREA-to-ACT	40	40	1 tCK
PREA-to-PRE	2	2	1 tCK
PREA-to-PREA	2	2	1 tCK
PREA-to-AREF	40	40	1 tCK
PREA-to-SRE	sys	sys	sys
PREA-to-PDE	sys	sys	sys
PREA-to-MRS	sys	sys	sys
AREF-to-DES	1 tCK	1 tCK	1 tCK
AREF-to-NOP	1 tCK	1 tCK	1 tCK
AREF-to-BST	NA dep	NA dep	NA dep
AREF-to-RD	NA	NA	1 tCK
AREF-to-WR	NA	NA	1 tCK
AREF-to-WRA	NA	NA	1 tCK
AREF-to-PRE	708	708	1 tCK
AREF-to-PREA	708	708	1 tCK
AREF-to-RDA	NA	NA	1 tCK
AREF-to-SRE	708	708	sys
AREF-to-PDE	708	708	sys
AREF-to-MRS	708	708	sys
AREF-to-ACT	708	708	1 tCK
AREF-to-AREF	708	708	1 tCK
MRS-to-DES	1 tCK	1 tCK	1 tCK
MRS-to-NOP	1 tCK	1 tCK	1 tCK
MRS-to-BST	NA dep	NA dep	NA dep
MRS-to-RD	34	34	sys
MRS-to-WR	34	34	sys
MRS-to-WRA	34	34	sys
MRS-to-ACT	34	34	sys
MRS-to-PRE	34	34	sys
MRS-to-PREA	34	34	sys
MRS-to-RDA	34	34	sys
MRS-to-AREF	34	34	sys
MRS-to-SRE	34	34	sys
MRS-to-PDE	34	34	sys
MRS-to-MRS	34	34	sys
SRE-to-DES	sys	sys	sys
SRE-to-NOP	sys	sys	sys
SRE-to-BST	sys	sys	sys
SRE-to-RD	sys	sys	sys
SRE-to-WR	sys	sys	sys

Command pair	same bank	diff bank	diff rank
SRE-to-WRA	sys	sys	sys
SRE-to-ACT	sys	sys	sys
SRE-to-PRE	sys	sys	sys
SRE-to-PREA	sys	sys	sys
SRE-to-RDA	sys	sys	sys
SRE-to-AREF	sys	sys	sys
SRE-to-SRE	sys	sys	sys
SRE-to-SRX	sys	sys	sys
SRE-to-PDE	sys	sys	sys
SRE-to-MRS	sys	sys	sys
SRX-to-DES	sys	sys	sys
SRX-to-NOP	sys	sys	sys
SRX-to-BST	sys	sys	sys
SRX-to-RD	720	720	sys
SRX-to-WR	720	720	sys
SRX-to-WRA	sys	sys	sys
SRX-to-ACT	720	720	sys
SRX-to-PRE	sys	sys	sys
SRX-to-PREA	sys	sys	sys
SRX-to-RDA	sys	sys	sys
SRX-to-AREF	sys	sys	sys
SRX-to-SRE	sys	sys	sys
SRX-to-PDE	sys	sys	sys
SRX-to-MRS	sys	sys	sys
PDE-to-DES	sys	sys	sys
PDE-to-NOP	sys	sys	sys
PDE-to-BST	sys	sys	sys
PDE-to-RD	sys	sys	sys
PDE-to-WR	sys	sys	sys
PDE-to-WRA	sys	sys	sys
PDE-to-ACT	sys	sys	sys
PDE-to-PRE	sys	sys	sys
PDE-to-PREA	sys	sys	sys
PDE-to-RDA	sys	sys	sys
PDE-to-AREF	sys	sys	sys
PDE-to-SRE	sys	sys	sys
PDE-to-PDE	sys	sys	sys
PDE-to-MRS	sys	sys	sys
PDE-to-PDX	sys	sys	sys
PDX-to-DES	sys	sys	sys
PDX-to-NOP	sys	sys	sys
PDX-to-BST	sys	sys	sys
PDX-to-RD	18	18	sys
PDX-to-WR	18	18	sys
PDX-to-WRA	sys	sys	sys
PDX-to-ACT	18	18	sys
PDX-to-PRE	sys	sys	sys
PDX-to-PREA	sys	sys	sys
PDX-to-RDA	sys	sys	sys
PDX-to-AREF	sys	sys	sys
PDX-to-SRE	sys	sys	sys
PDX-to-MRS	sys	sys	sys
PDX-to-PDE	sys	sys	sys